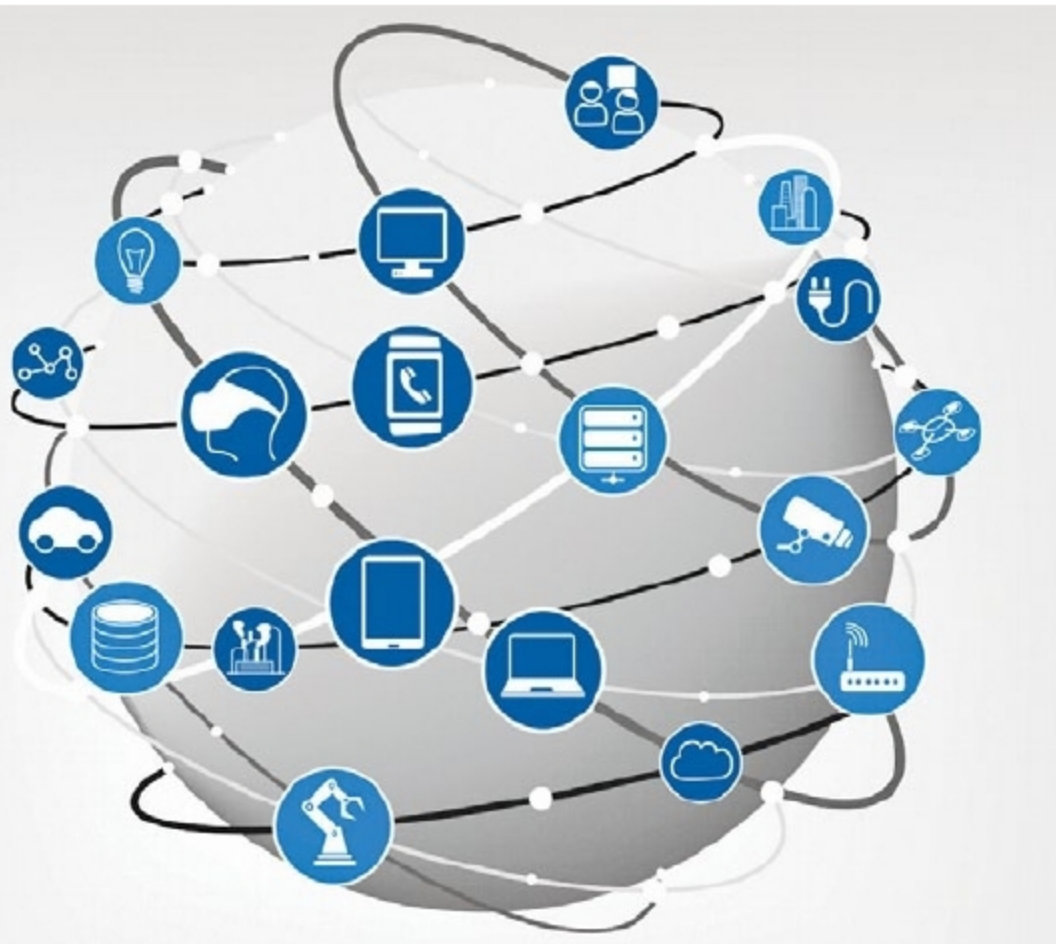


COMMUNICATION

And INTERNET TECHNOLOGY: Interconnecting Networks



Interconnection networks are high-speed computer networks. These are connections between nodes where each node can be a single processor or a group of processors or memory modules. These connections carry or transport data from one processor to another, or from the processor to the memory so that the task is broken down and computed in parallel



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Individual networks have their own coverage area. The need was felt to extend the coverage area of networks by inter-networking. In particular, the need was felt to achieve global connectivity to take full advantage of networking, namely, global linking, resource sharing and load sharing.

There were two challenges in inter-networking for global connectivity. Different countries have their own policies, programmes, priorities, choices and preferences. As such, it was impossible to use jurisdictional geography of one country by another to implement physical interconnecting networks.

Moreover, individual networks of different countries are designed based on different protocols. Protocol of a network is a set of rules for the functional operation and connection of nodes, devices and computers to the network. However, resolving the incompatibilities of individual networks for inter-networking was a challenge.

Gateways as devices are one set of tools for resolving the issue. Protocol converters are typical examples of the simplest gateways. For example, let us assume three individual networks, namely, a, b and c, are designed for three different protocols, namely, A, B and C,